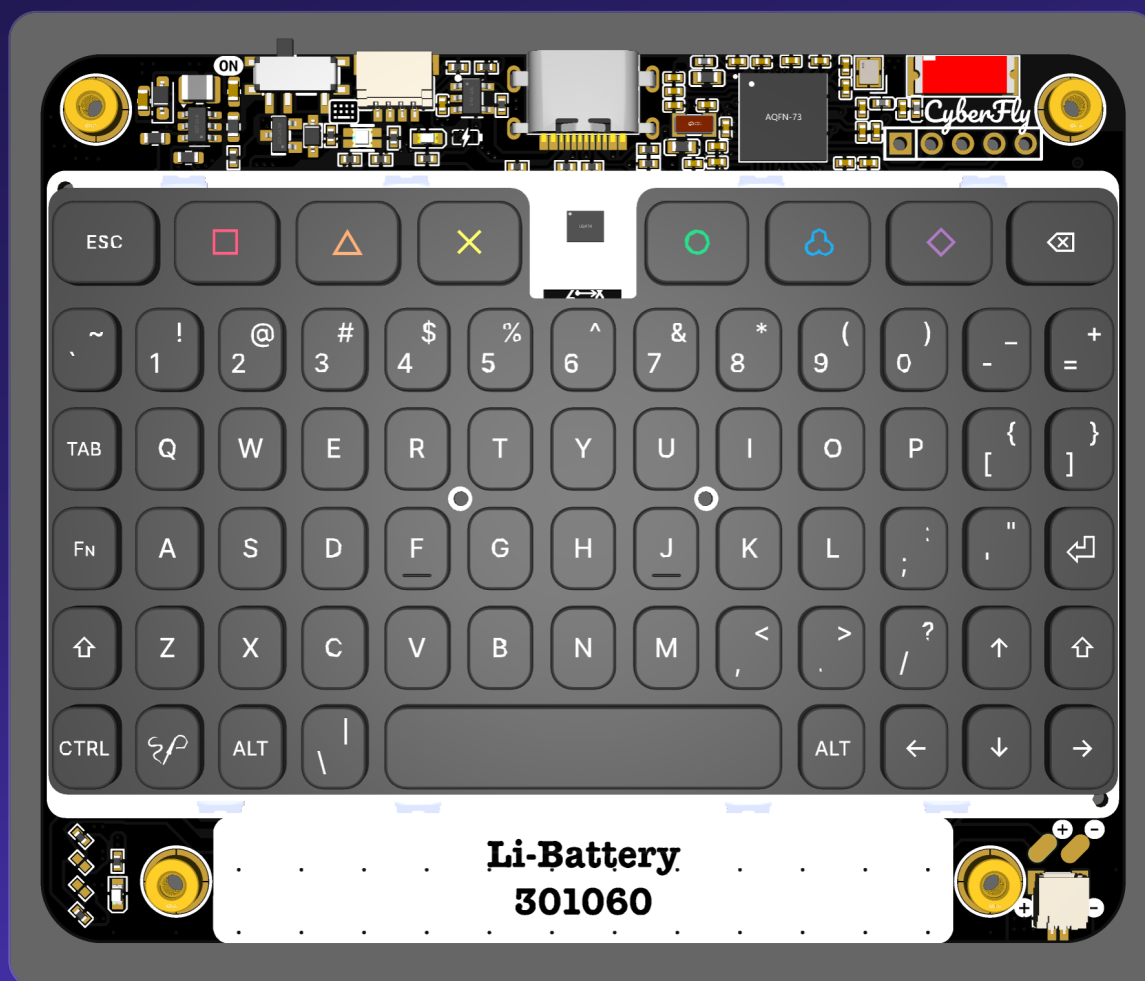


CYBERFLY · NRF52840 · ZMK

TypixNode

BLE + USB 双模便携键盘 · 使用说明书

BLE + USB Dual-mode Portable Keyboard · User Manual | 中文 / English



MCU

nRF52840

无线 / LINK

BLE + USB

蓝牙 / BT

6 设备

飞鼠 / AIR

6 轴 IMU

固件 / FW

ZMK

TypixNode 是一款卡片大小的 BLE + USB 双模 便携键盘，基于 nRF52840 (Nice!Nano v2 引脚兼容) + ZMK 固件。支持最多 6 台蓝牙设备 切换，板载 QMI8658A 6 轴 IMU 可作空间飞鼠，带 RGB 状态灯与白色背光。BLE 下 macOS / iOS 原生显示电量。约 69 键，含一个 Fn 功能层。

1 产品概览

产品名称	TypixNode (CyberFly 硬件)
类型	BLE + USB 双模便携键盘
主控	nRF52840 (Nice!Nano v2 兼容)
固件	ZMK (定制分支)
蓝牙	BLE 5.0, 最多 6 台设备 配对切换
飞鼠	QMI8658A 6 轴 IMU (加速度计 + 陀螺仪)
背光	白色, 仅 USB 通电可用
电池	锂聚合物 LP301060 (约 250 mAh)



透明亚克力 + 白色背光实拍；底部红 / 白 LED 为充电 / USB 指示

2 指示灯一览

位置	灯	含义
键盘板上方	● RGB 灯	工作状态 (开机动画、USB 循环、BLE 心跳、飞鼠切换)
RGB 旁	● 红灯	充电中常亮，充满自动熄灭
键盘板下方	○ 白灯	USB 5V 供电指示，插线就亮

RGB 状态详解

场景	效果
开机	红 → 绿 → 蓝 顺序闪一次后熄灭 (启动动画)
USB 已连接	持续循环变换 (三色轮换)
BLE 待机	● 蓝色双闪, 每 5 秒一次心跳, 代表随时可用
飞鼠切换	OFF=● 红闪 / M1=● 绿闪 / M2=● 蓝闪 (双闪后回归原状态)

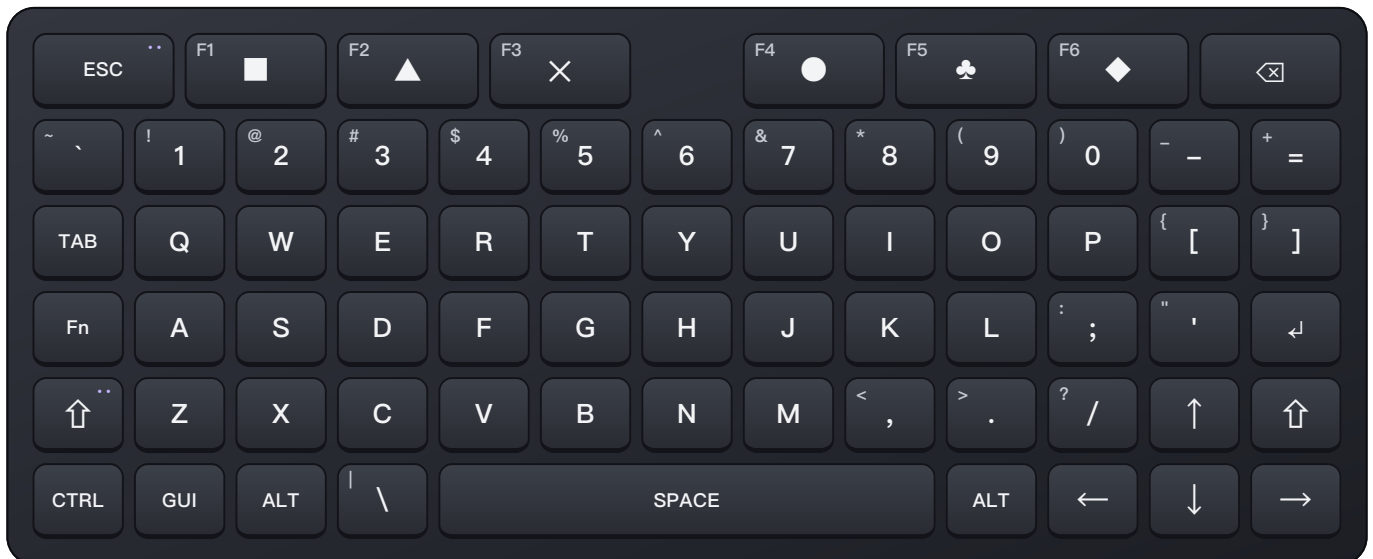
3 工作模式

模式	连接	背光	飞鼠
USB	USB-C 线	可用 (Fn+空格调节)	可用
BLE	蓝牙	自动关闭 (省电)	需手动开启

背光仅 USB 通电时可用，拔线自动熄灭 (`AUTO_OFF_USB`)；空闲也会自动熄灭 (`AUTO_OFF_IDLE`)。

4 键位图 · 基础层 (Base)

下图为按 PCB 丝印呈现的物理键位（标准 QWERTY）。左上角小字为 **Shift** 上档符号；标记 `..` 的键支持双击（见第 7 节）。



■ ▲ × ● ♣ ◆ = 顶部 6 个彩色符号键 = `F1-F6` `..` = 双击有隐藏功能 左上小字 = Shift 上档

5 键位图 · Fn 层

按住左侧 **Fn** 键 (CapsLock 位) 时，高亮键生效，灰色为透传。 蓝色 = 蓝牙/输出相关， 紫色 = 其它功能。



按住 Fn +	功能
Esc	切换 USB / 蓝牙 输出 (<code>OUT_T0G</code>)
■ ▲ × ● ♣ ◆	切换到蓝牙设备 1—6 (<code>BT_SEL 0-5</code>)
Enter	清除当前蓝牙设备的配对 (<code>BT_CLR</code>)，用于重新配对该槽位
<code>1 ~ 0 / - / =</code>	F1 ~ F10 / F11 / F12
⌫ Backspace	Delete
↑ / ↓ / ← / →	PgUp / PgDn / Home / End
空格 Space	循环切换背光亮度 (<code>BL_CYCLE</code>)
左 Alt / 右 Alt	开 / 关 6 轴飞鼠 (OFF → M1 → M2 循环)
右 Shift (按住)	软关机 (<code>soft_off</code>)
]	外接电源开关 (<code>EP_T0G</code>)
Q / A / M	截图宏 / 全选复制宏 / <code>() => {}</code> 宏
C	重置全部设置 (清 NVS) 并重启 [★]
B	进入 Bootloader (UF2 刷机)

6 蓝牙多设备连接

固件支持 最多 6 台蓝牙设备 (`CONFIG_BT_MAX_PAIRED=6`)，通过 Fn + 顶部符号键在 6 个槽位间切换。

操作	按键
选择 / 切换到设备 1—6	Fn + ■ / ▲ / × / ● / ♣ / ◆
清除当前设备配对 (重新配对该槽位)	Fn + Enter
切换 USB / 蓝牙 输出	Fn + Esc
重置全部设置 (清空所有配对 + 重启)	Fn + C [★]

配对流程

- 1. Fn + 符号键 选一个空槽位（如设备 1 = Fn + █）。
- 2. 键盘进入广播，在主机蓝牙设置里搜索 TypixNode KBD 并配对。
- 3. 连接后，系统「我的设备」即显示电量。
- 4. 换主机：Fn + 另一个符号键 切到别的槽位再配对即可——各设备互不覆盖。

两种清除的区别：

- Fn + Enter = 只清当前这台的配对（其它设备保留）。
- Fn + C = 核弹级，清空全部配对、设备名缓存等所有设置并重启，配对全乱时用它。

电量显示

- macOS：系统设置 → 蓝牙 → 「我的设备」卡片。
- iOS / iPadOS：桌面加「电池」小组件。
- 协议：标准 BLE Battery Service (0x180F)。

7 隐藏技巧：双击 / Combo / 宏

双击 (Tap-dance)

键	单击	双击
Esc	Esc	Caps Word（一次性大写，遇空格退出）
左 Shift	Shift	锁定 Caps Lock

Combo 组合键 (同时按)

同按	触发	判定窗口
J + K	Esc	50 ms
D + F	Tab	50 ms
F + J	Enter	50 ms
J + K + L	Caps Word	50 ms
Fn + 右 Shift	开 / 关飞鼠	75 ms

宏 (Fn 层)

键	宏效果
Fn + Q	截屏 (⌘ + Shift + 4)
Fn + A	全选 + 复制 (Ctrl+A → 50ms → Ctrl+C)
Fn + M	输入 () => {} 箭头函数

关于 Fn + 右 Shift：75ms 内同时按下触发飞鼠切换；分前后按则走 Fn 层的软关机。想关机就慢点按，想切飞鼠就同时按下。

8 6 轴飞鼠 (Air Mouse)

板载 QMI8658A 6 轴 IMU（加速度计 + 陀螺仪，I²C）。三档循环切换：

模式	算法	状态灯	风格
OFF	关闭，IMU 进省电	● 红闪	纯键盘
M1	Kalman 滤波 + 状态空间融合	● 绿闪	稳
M2	加速度倾角映射（简易回退）	● 蓝闪	跟手

三种切换方式：Fn + 右 Shift（75ms 同按）、Fn + 左 Alt、Fn + 右 Alt。切换时 RGB 对应颜色双闪确认。

Smart Space：飞鼠开启后空格键变鼠标

位置	OFF	ON (M1/M2)
左空格	空格	鼠标左键
中空格	空格	空格（保持）
右空格	空格	鼠标右键

提示：飞鼠仍在调优（漂移滤波、加速度曲线），稳定版前仅作体验。

9 背光

- 白色背光，PWM 经 AP3032 升压驱动（P0.15），开机默认关闭，步进 20%。
- Fn + 空格 循环切换亮度。
- 仅 USB 通电 时可用；拔线或空闲自动熄灭（省电）。

10 电源 / 充电 / 开关

- 充电口：USB-C，板载充电管理；● 红灯充电中常亮，满电自动灭。
- ○ 白灯：USB 5V 供电指示，插线就亮。
- 硬件电源拨动开关：ON = 电池供电、BLE 正常；OFF = 物理断电池，零功耗长期存放。
- Fn + 右 Shift（按住）软关机；Fn +] 切换外接电源（EP_T0G）。
- 配对信息存 flash NVS，断电不丢。BLE 待机 > 48 小时。

11 Bootloader / 固件升级

进入 Bootloader 的两种方式

1. Fn + B（正常工作时）。
2. 板载 reset 按钮双击（固件挂了或 Fn+B 失效时）。

进入后：板子主 LED 慢速呼吸，出现名为 **NICENANO** 的 FAT 虚拟 U 盘，把 **.uf2** 拖进去即自动重启刷新。

本地编译固件

```
cd cyberfly_zmk
rm -rf build && .venv/bin/west build -s app -p \
  -b nice_nano/nrf52840/zmk -- -DSHIELD=cyberfly
# 产物：build/zephyr/zmk.uf2
```

固件栈

- Bootloader: Adafruit nRF52 UF2 **v0.10.0** (nice!nano fork)
- SoftDevice: Nordic **S140 v6.1.1** (BLE 协议栈)
- 应用层: ZMK 定制分支 (Smart Space、Toggle Mouse、RGB Status LED、Reset Settings 等自定义 behavior)

12 硬件规格

项目	参数
主控	nRF52840 (Nice!Nano v2 兼容)
无线	BLE 5.0 (S140 v6.1.1) + USB HID
键盘矩阵	6 行 × 13 列, 二极管方向 row2col
去抖	按下 10 ms / 释放 10 ms
IMU	QMI8658A 6 轴 (I ² C)
外部 I ² C / Qwiic	JST-SH 1.0mm 4-pin (GND / 3V3 / SDA / SCL), 兼容 Qwiic / STEMMA QT
背光	PWM 驱动 AP3032 升压 LED driver @ P0.15
RGB 状态灯	PWM1 × 3 通道 (R / G / B)
外部电源控制	P0.13 → LDO CE (Fn +] 切换)
充电	USB-C, 板载管理, 红灯指示
蓝牙绑定	最多 6 台设备 (CONFIG_BT_MAX_PAIRED=6)
电池	LP301060 锂聚合物 (约 250 mAh)
USB 标识	VID 0x1209 / PID 0x0001, TypixNode Keyboard
BLE 设备名	TypixNode KBD

13 常见问题 (FAQ)

Q: 怎么把键盘连到第二台、第三台设备?

A: Fn + 不同的符号键 切到别的蓝牙槽位再配对即可, 最多 6 台, 互不覆盖。不再需要每次清除。

Q: 某台主机配对抽风 / 连不上?

A: 先 Fn + Enter 清掉当前这台的配对重配; 还不行用 Fn + C 重置全部设置。

Q: 背光不亮?

A: 背光只在 USB 通电时可用, BLE 模式自动关闭。Fn + 空格 调亮度。

Q: 电量显示不准?

A: 首次连接等 1 分钟左右让 BLE 电池服务同步。

Q: Fn + B 后没出现 NICENANO 盘?

A: USB 线可能只通电不走数据, 换一根 data 线, 或双击 reset 键。

Q: 按 Fn + 右 Shift 有关机有时切飞鼠?

A: 75ms 同按切飞鼠; 分前后按走软关机。想关机慢点按, 想切飞鼠同时按。

Q: 长期不用怎么处理?

A: 硬件开关拨 OFF 物理断电, 配对信息保留在 flash 不丢失。

The TypixNode is a card-sized **BLE + USB dual-mode** portable keyboard built on an **nRF52840** (Nice!Nano v2 compatible) running **ZMK** firmware. It pairs with **up to 6 Bluetooth devices**, carries a **QMI8658A 6-axis IMU** for an air-mouse, and has an RGB status LED plus white backlight. On BLE it reports battery natively to macOS / iOS. About **69 keys** with one **Fn layer**.

1 Overview

Product	TypixNode (CyberFly hardware)
Type	BLE + USB dual-mode keyboard
MCU	nRF52840 (Nice!Nano v2 compatible)
Firmware	ZMK (custom fork)
Bluetooth	BLE 5.0, up to 6 paired devices
Air mouse	QMI8658A 6-axis IMU
Backlight	White, USB-power only
Battery	LiPo LP301060 (~250 mAh)



Clear acrylic with white backlight; the red / white LEDs are charge / USB indicators

2 Indicator LEDs

Location	LED	Meaning
Top of board	● RGB	Status (boot animation, USB cycle, BLE heartbeat, mouse toggle)
Next to RGB	● Red	Solid while charging, off when full
Bottom of board	○ White	USB 5V power present

RGB status details

Scene	Effect
Boot	Red → Green → Blue flash once, then off
USB connected	Continuous colour cycle
BLE standby	● Blue double-blink every 5 s (heartbeat)
Mouse toggle	OFF=● red / M1=● green / M2=● blue blink

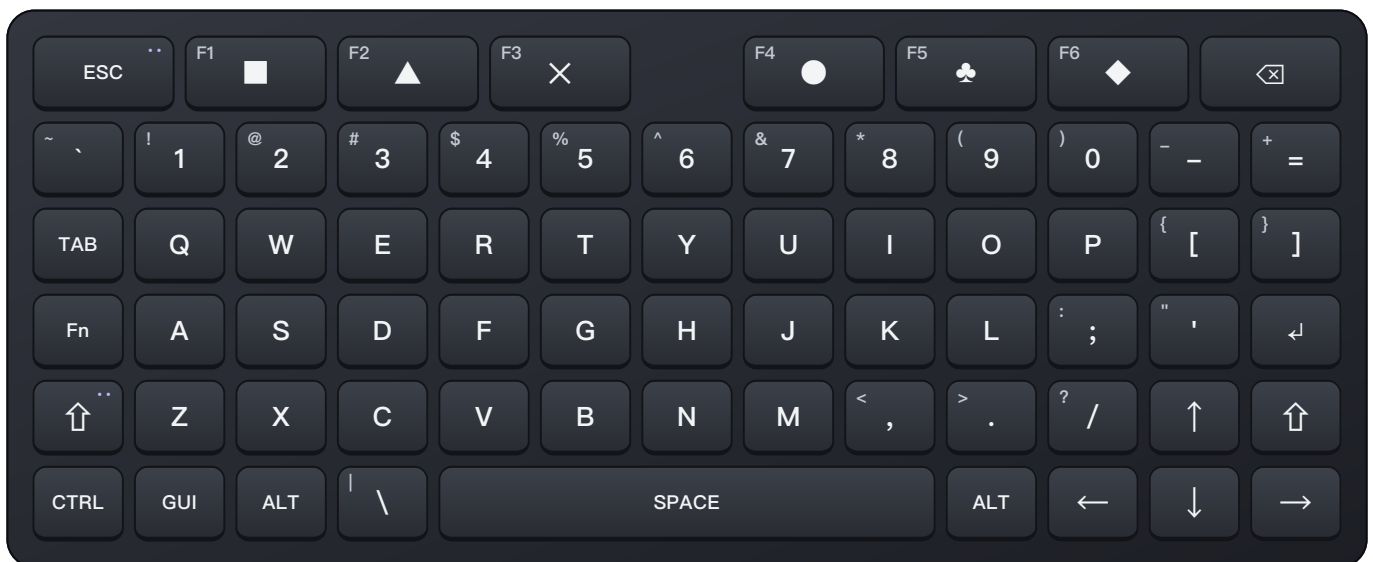
3 Operating Modes

Mode	Link	Backlight	Air mouse
USB	USB-C cable	Available (Fn+Space)	Available
BLE	Bluetooth	Auto-off (saving)	Manual enable

Backlight is USB-power only — it turns off when unplugged (`AUTO_OFF_USB`) and when idle (`AUTO_OFF_IDLE`).

4 Keymap · Base Layer

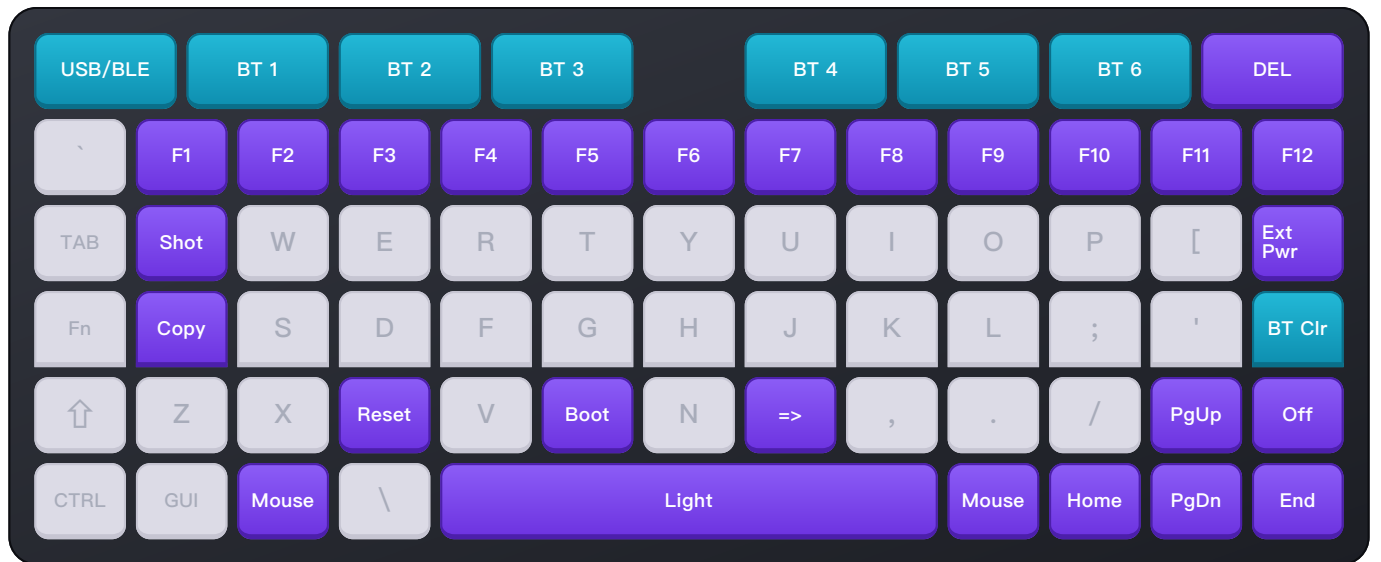
The diagram shows the **physical layout** as silk-screened (standard QWERTY). Small top-left = **Shift** symbol; keys marked `..` have a **double-tap** action (see §7).



■ ▲ × ● ♣ ◆ = six coloured symbol keys = `F1–F6` `..` = double-tap action top-left small = Shift

5 Keymap · Fn Layer

Hold the left **Fn** key (CapsLock position). Highlighted keys change; grey ones pass through. **Cyan** = Bluetooth / output, **violet** = other functions.



Hold Fn +	Function
Esc	Toggle USB / BLE output (<code>OUT_T0G</code>)
■ ▲ × ● ♣ ◆	Switch to Bluetooth device 1–6 (<code>BT_SEL 0–5</code>)
Enter	Clear the current device's bond (<code>BT_CLR</code>) to re-pair that slot
<code>1 – 0 / – / =</code>	F1–F10 / F11 / F12
⌫ Backspace	Delete
↑ / ↓ / ← / →	PgUp / PgDn / Home / End
Space	Cycle backlight (<code>BL_CYCLE</code>)
Left Alt / Right Alt	Toggle 6-axis air mouse (OFF → M1 → M2)
Right Shift (hold)	Soft off (<code>soft_off</code>)
]	External power toggle (<code>EP_T0G</code>)
Q / A / M	Screenshot / select-all+copy / <code>() => {}</code> macro
C	Reset all settings (erase NVS) and reboot ⭐
B	Enter bootloader (UF2)

6 Multi-device Bluetooth

The firmware supports **up to 6 paired devices** (`CONFIG_BT_MAX_PAIRED=6`), switched with Fn + the top symbol keys.

Action	Keys
Select / switch to device 1–6	Fn + ■ / ▲ / × / ● / ♣ / ♦
Clear current device's bond (re-pair slot)	Fn + Enter
Toggle USB / BLE output	Fn + Esc
Reset all settings (wipe all bonds + reboot)	Fn + C 

Pairing

1. Fn + **symbol key** to pick an empty slot (e.g. device 1 = Fn + ■).
2. The keyboard advertises; find **TypixNode KBD** in the host's Bluetooth settings and pair.
3. Battery then shows up in the host's device card.
4. To use another host, Fn + a **different symbol key** and pair — **slots don't overwrite each other**.

Two kinds of clearing:

- Fn + Enter = clear only the **current** device (others kept).
- Fn + C = **nuclear**: wipe **all** bonds, name cache and settings, then reboot. Use when pairing is fully stuck.

Battery display

- macOS: System Settings → Bluetooth → device card.
- iOS / iPadOS: add the Batteries widget.
- Protocol: standard **BLE Battery Service (0x180F)**.

7 Hidden Tricks: Double-tap / Combos / Macros

Tap-dance

Key	Single	Double
Esc	Esc	Caps Word (one-shot caps, ends on space)
Left Shift	Shift	Caps Lock

Combos (press together)

Together	Sends	Window
J + K	Esc	50 ms
D + F	Tab	50 ms
F + J	Enter	50 ms
J + K + L	Caps Word	50 ms
Fn + Right Shift	Toggle air mouse	75 ms

Macros (Fn layer)

Key	Macro
Fn + Q	Screenshot (⌘ + Shift + 4)
Fn + A	Select all + copy (Ctrl+A → 50ms → Ctrl+C)
Fn + M	Type <code>() => {}</code>

About Fn + Right Shift: pressed **together** within 75 ms it toggles the air mouse; pressed **in sequence** it triggers **soft-off** on the Fn layer.

8 6-axis Air Mouse

On-board QMI8658A 6-axis IMU (accel + gyro, I²C). Three modes, cycled:

Mode	Algorithm	LED	Feel
OFF	Off, IMU sleeps	● red	keyboard only
M1	Kalman filter + state-space fusion	● green	steady
M2	Accelerometer tilt-to-velocity (fallback)	● blue	responsive

Three ways to toggle: Fn + Right Shift (within 75 ms), Fn + Left Alt, Fn + Right Alt. The RGB blinks the matching colour to confirm.

Smart Space: spacebar becomes mouse buttons when the mouse is on

Position	OFF	ON (M1/M2)
Left space	Space	Left click
Middle space	Space	Space (kept)
Right space	Space	Right click

Note: the air mouse is still being tuned (drift filtering, accel curve) — experimental for now.

9 Backlight

- White backlight via an **AP3032 boost driver** (P0.15), **off** at boot, 20% steps.
- **Fn + Space** cycles brightness.
- Only available on **USB power**; turns off when unplugged or idle.

10 Power / Charging / Switch

- Charge port: **USB-C** with on-board charging; ● red LED solid while charging, off when full.
- ○ White LED = USB 5V present.
- Hardware slide switch: **ON** = battery on, BLE active; **OFF** = physically disconnect battery for zero-drain storage.
- **Fn + Right Shift** (hold) soft-off; **Fn +]** toggles external power (`EP_TOG`).
- Pairing data lives in flash NVS — **kept across power loss**. BLE standby > 48 h.

11 Bootloader / Firmware

Two ways into the bootloader

1. **Fn + B** (while running).
2. **Double-tap the reset button** (if firmware is stuck).

Then the main LED breathes slowly and a FAT drive named `NICENANO` appears — drop a `.uf2` onto it to flash and auto-reboot.

Build locally

```
cd cyberfly_zmk
rm -rf build && .venv/bin/west build -s app -p \
  -b nice_nano/nrf52840/zmk -- -DSHIELD=cyberfly
# output: build/zephyr/zmk.uf2
```

Firmware stack

- Bootloader: Adafruit nRF52 UF2 **v0.10.0** (nice!nano fork)
- SoftDevice: Nordic **S140 v6.1.1**
- App: custom ZMK fork (Smart Space, Toggle Mouse, RGB Status LED, Reset Settings behaviours)

12 Hardware Specifications

Item	Spec
MCU	nRF52840 (Nice!Nano v2 compatible)
Wireless	BLE 5.0 (S140 v6.1.1) + USB HID
Matrix	6 rows × 13 cols, row2col diodes
Debounce	press 10 ms / release 10 ms
IMU	QMI8658A 6-axis (I ² C)
External I ² C / Qwiic	JST-SH 1.0mm 4-pin (GND / 3V3 / SDA / SCL), Qwiic / STEMMA QT compatible
Backlight	PWM AP3032 boost LED driver @ P0.15
RGB status LED	PWM1 × 3 channels (R / G / B)
External power	P0.13 → LDO CE (<code>Fn + J</code>)
Charging	USB-C, on-board, red LED
BT bonding	up to 6 devices (<code>CONFIG_BT_MAX_PAIRED=6</code>)
Battery	LP301060 LiPo (~250 mAh)
USB IDs	VID <code>0x1209</code> / PID <code>0x0001</code> , TypixNode Keyboard
BLE name	<code>TypixNode KBD</code>

13 FAQ

Q: How do I connect to a 2nd / 3rd device?

A: `Fn + a different symbol key` selects another Bluetooth slot — pair there. Up to 6, none overwrite each other.

Q: A host won't connect / pairing is flaky?

A: `Fn + Enter` clears the **current** device's bond and re-pair; if still stuck, `Fn + C` resets everything.

Q: Backlight is off?

A: It's USB-power only and auto-off on BLE. `Fn + Space` adjusts brightness.

Q: Battery % looks wrong?

A: Wait ~1 min after connecting for the BLE battery service to sync.

Q: No NICENANO drive after `Fn + B`?

A: The cable may be charge-only — use a data cable, or double-tap reset.

Q: `Fn + Right Shift` sometimes powers off, sometimes toggles the mouse?

A: Together within 75 ms = mouse; in sequence = soft-off.

Q: Long-term storage?

A: Flip the hardware switch OFF — pairing data is retained in flash.